Justin Hsu

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Education

Georgia Institute of Technology — Atlanta, GA
Master of Science in Electrical Engineering — 2026-2028

University of California, Davis — Davis, CA
Bachelor of Science in Electrical Engineering — 2022-2026

Experience

RTL Design Engineer Intern — Acton, MA
Lotus Communication Systems — June 2025-September 2025

- Designed and implemented a high-performance digital Finite Impulse Response (FIR) filter and Block Up/Down Converter (BUDC) to optimize signal processing efficiency for Xilinx FPGA deployment using SystemVerilog.
- Achieved a 19% reduction in hardware area and improved operational clock frequency to 200 MHz by leveraging Vivado and MATLAB to execute Power, Performance, and Area (PPA) analysis, rewriting critical RTL blocks to minimize multiply-and-accumulate (MAC) calculations.
- Validated system functionality through rigorous RTL simulation and timing closure verification, successfully integrating the BUDC block into the larger communication system pipeline to ensure error-free up/down-conversion.

PCB Layout Intern — Acton, MA
Lotus Communication Systems — June 2024-September 2024

- Designed and implemented a complex mixed-signal PCB layout for a gigahertz-frequency Block Up/Down Converter (BUDC) with the goal of isolating high-frequency RF paths from digital control circuitry.
- Achieved 40 dB of noise isolation and met strict 50 ohm controlled impedance requirements across all critical paths using Siemens PADS and Cadence Allegro to optimize differential pair routing, component placement, and board area.
- Validated signal integrity and layout performance through post-route parasitic extraction and simulation, ensuring successful hardware integration and error-free operation of the converter within the larger RF front-end system.

Projects

Handwritten Number CNN Accelerator — April 2026-June 2026

- Architected a 2-layer CNN accelerator on an Altera DE1-SoC FPGA, performing a comprehensive microarchitectural trade-off analysis across weight, input, output, and row-stationary systolic array dataflows to optimize MAC computational density and memory bandwidth.
- Achieved an operational clock frequency of 100 MHz and reduced power consumption by 20% under Skywater 130nm PDK by implementing clock gating and multi-voltage domain floorplanning via Synopsys Design Compiler and Cadence Innovus.
- Verified the physical design and microarchitecture through complete RTL-to-GDSII physical verification and post-layout timing closure, ensuring successful hardware-software co-design integration for the larger neural network inference system.

Subthreshold Minimum Energy 32-Bit Adder — January 2026-March 2026

- Engineered a low-power 32-bit Square Root Carry Select Adder (SRCSA) optimized for subthreshold operation, aiming to maximize energy efficiency while maintaining high-speed computational throughput.
- Achieved a clock frequency of 100 MHz at an ultra-low supply voltage of Vdd = 0.32V by leveraging Cadence Virtuoso to design a custom cell library utilizing even/odd mirror adder topologies and a strategic mix of multi-threshold devices (High-Vt, Normal-Vt, Low-Vt).
- Verified the entire layout design by conducting strict 45nm CMOS DRC/LVS checks, performing post-layout parasitic extraction (PEX), and utilizing Cadence Spectre and Xcelium to characterize static leakage, Power-Delay Product (PDP), and Energy-Delay Product (EDP) to guarantee robust silicon performance.

Heterogeneous Multicore RISC-V Processor — January 2025-March 2026

- Designed a heterogeneous RISC-V multi-core architecture in SystemVerilog, integrating a deeply pipelined, high-performance "Big" core alongside an energy-efficient "LITTLE" core to maximize instruction throughput under strict thermal design power constraints.
- Achieved an operational frequency of 400 MHz and a 30% increase in energy efficiency using Synopsys Design Compiler and Cadence Innovus to optimize pipeline register placement, clock tree synthesis (CTS), multi-voltage floorplanning, and routing congestion.
- Verified design functionality, timing closure, and protocol compliance by authoring self-checking, constrained-random testbenches in Cadence Xcelium to rigorously stress the hardware-enforced cache coherence protocol and ensure flawless multi-core synchronization.

Skills

Languages: SystemVerilog, Verilog, C, Python, TCL, MATLAB
ASIC/EDA Tools: Cadence Spectre, Xcelium, Innovus, Virtuoso, Allegro, Synopsys Design Compiler, Siemens PADS, Vivado, Quartus, Verilator, MAGIC, Yosys, KLayout
Technical Competencies: STA, Layout, PPA Analysis, Logic Design, Low Power Design, HW-SW Co-design, Microsoldering